MIL-P-13234C
10 May 1977
SUPERSEDING
MIL-P-13234B
28 May 1971

MILITARY SPECIFICATION

PUMPS, RECIPROCATING, HAND-DRIVEN

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers horizontal, double-acting, reciprocating pumps rated in gallons per cycle (gpc).

1.2 Classification. The pumps shall be of the following types and sizes, as specified (see 6.2):

Type I - Double lever.
  Size 1 - 0.60 gpc.
  Size 2 - 1.30 gpc.
Type II - Single lever, 0.60 gpc.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

FSC 4320

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

QQ-S-781	- Strapping, Steel, and Seals.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-P-40	- Packaging and Packing of Hand Tools.

MILITARY

MIL-P-116	- Preservation-Packaging, Methods of.
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-T-704	- Treatment and Painting of Materiel.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 47 - Malleable Iron Castings.
A 48 - Gray Iron Castings.
B 21 - Naval Brass Rod, Bar, and Shapes.
B 62 - Composition Bronze or Ounce Metal Castings.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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NATIONAL BUREAU OF STANDARDS (NBS) .

Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Traffic Department, 1616 P Street, NW, Washington, DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification Rules.

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Description. The pumps shall be the horizontal, single-cylinder, reciprocating, double-acting, hand-driven type. Each pump shall be mounted on a hardwood base and shall be furnished with adapters and caps necessary for operation as specified herein.

3.2 First article (preproduction model). The contractor shall furnish one or more pumps as specified (see 6.2) for examination and testing within the time frame specified (see 6.2) to prove prior to starting production that his production methods and choice of design detail will produce pumps that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

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3.3 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Gaskets. Gaskets shall be provided for all jointed surfaces subject to pressure.

3.4 Screw threads. Screw threads shall conform to NBS Handbook H28.

3.5 Ease of maintenance. All assemblies and installed attachments shall be accessible for servicing, repair, and replacement without removing the pump body from its base.

3.6 Pump body, air chamber, cylinder heads, and piston plates. The pump body, air chamber, cylinder heads, and piston plates shall be gray iron castings conforming to ASTM A 48, Class 30. The pump body shall be provided with internally threaded priming and drain openings. The priming opening shall be .750 - 10UNC and the drain opening shall be .375 - 16UNC.

3.7 Hydrostatic pressure. The pump body (including air chamber) and valves shall not leak, fail to operate, or show permanent deformation when subjected to an internal hydrostatic pressure of not less than 110 pounds per square inch (psi).

3.8 Valves, valve seats, and packing glands. Valves, valve seats, and packing glands shall be brass castings conforming to ASTM B 62.

3.9 Lever arms. Lever arms shall be detachable and shall be malleable cast iron conforming to ASTM A 47, grade optional.

3.10 Stuffing box and piston packing. Stuffing box and piston packing shall be synthetic rubber-frictioned cotton fabric layers and shall not be damaged by contact with petroleum products.

3.11 Priming plugs and drain plugs. Plugs shall be brass conforming to ASTM B 21, Alloy 482. Priming plugs shall be externally threaded .750 - 10UNC. Drain plugs shall be externally threaded .375 - 16UNC.

3.12 Suction and discharge adapters and caps. The suction and discharge openings of each pump shall be provided with adapters and caps. The caps shall withstand repeated use and shall be attachable and detachable by hand. The caps shall be captive chained to the pump. The adapters and caps shall be brass castings conforming to ASTM B 62.

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3.13 Pump base. The pumps shall be mounted on a hardwood base having dimensions as shown on Figure 1 or 2, as applicable. The base shall have a galvanized steel eye bolt with an inside diameter of not less than 1 inch attached near each corner. Wood shall be fully seasoned and free of loose knots, worm holes, checks, and splits.

3.14 Priming. When operating the pump at a static suction lift equivalent to 20 feet at standard sea level atmospheric conditions of 29.92 inches of mercury and a specific weight of water of 62.3 pounds per cubic foot (lb/cu ft) at a temperature of 68° F, the pumps shall prime in not more than 100 strokes. After priming, the pump shall operate without binding or sticking and shall deliver a steady liquid flow after displacing 1.0 cubic feet of fluid (air and water mixture).

3.15 Identification marking. The pump shall be identified in accordance with MIL-STD-130. The marking shall be applied to the pump on plates conforming to MIL-P-514, Type I, Style 1, Composition C, of Type I, Grade A, Class 1 material. Each plate shall be attached by screws, bolts, or rivets in a location where the plate will be both visible and legible.

3.16 Treatment and painting. The portions of the pump normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, Type A, color as specified (see 6.2). Nonpermissible galvanic couples (dissimilar metals) as defined in MIL-T-704 shall be insulated as specified in MIL-T-704.

3.17 Type I, double-lever pump. The double-lever pump shall deliver 0.60 gpc for Size 1 and 1.30 gpc for Size 2 at a static suction lift equivalent to 20 feet of water at standard conditions (see 3.14) and a discharge head of not less than 20 feet of water. The pump shall conform to the dimensions shown on Figure 1. Suction and discharge openings shall be internally threaded. Piston rods may be either single or double ended. The pump shall be provided with two lever arms connected by a malleable cast-iron yoke conforming to ASTM A 47, grade optional. Pump bore, stroke, weight, suction, and discharge shall conform to Table I. Adapters and caps shall conform to Table II.

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TABLE I. Double-lever pump: bore, stroke, weight, suction and discharge.

Size	Capacity (gpc)	Bore (min.) (inches)	Stroke (min.) (inches)	Weight (max.) (pounds)	Suction and discharge (inches)
Size 1	0.60 ± 0.03	4	5-1/2	175	2 - 11-1/2 NPT
Size 2	1.30 ± 0.06	6	5-1/2	290	3 - 8 NPT

TABLE II. Adapters and caps for double-lever pumps.

Quantity	Part	Size and thread	
		0.60 gpc	1.30 gpc
2	Adapters	2 - 11-1/2 NPT external to 1-1/2 - 11-1/2 NPSH external.	3 - 8 NPT external to 2-1/2 - 7-1/2 NH external.
2	Caps	1-1/2 - 11-1/2 NPSH.	2-1/2 - 7-1/2 NH

3.18 Type II, single lever pump. The single-lever pump shall deliver 0.60 gpc, plus or minus 0.03 gallon, at a static suction lift equivalent to 20 feet of water at standard conditions (see 3.14) and a discharge head of not less than 30 feet of water. The pump shall conform to the dimensions shown on Figure 2. The pump shall have a bore of not less than 4 inches and a stroke of not less than 5-1/2 inches. The weight of the pump shall be not more than 150 pounds. The pump shall be provided with a single-lever arm. Suction and discharge openings shall be internally threaded 2 - 11-1/2 NPT. Adapters and caps shall conform to Table III.

TABLE III. Adapters and caps for single-lever pumps.

Quantity	Part	Size and thread
1	Adapter (discharge)	2 - 11-1/2 NPT external to 1 - 11-1/2 NPT external.
1	Adapter (suction)	2 - 11-1/2 NPT external to 1-1/2 - 11-1/2 NPT external.
1	Cap (discharge)	1 - 11-1/2 NPT.
1	Cap (suction)	1-1/2 - 11-1/2 NPT.

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3.19 Workmanship. All parts, components, and assemblies of the pump including castings, forgings, molded parts, machined surfaces, and welded parts shall be clean and free from sand, dirt, fins, pits, sprues, scale, flux, and other harmful extraneous material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of packaging (see 4.6).

4.3 Preproduction inspection.

4.3.1 Examination. The preproduction pump shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The preproduction pump shall be tested as specified in 4.5.2.1 and 4.5.2.2. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for tests shall be in accordance with MIL-STD-105.

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4.4.2 Examination. Each pump shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4.3 Tests.

4.4.3.1 Individual. Each pump shall be tested as specified in 4.5.2.1. Failure of the test shall be cause for rejection.

4.4.3.2 Samples. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2.2. AQL shall be 4.0 percent defective.

4.5 Inspection procedure.

4.5.1 Examination. The pump shall be examined as specified herein for the following major defects:

101. Dimensions not as specified.
102. Materials not as specified.
103. Assemblies and attachments not accessible.
104. Components missing or not as specified.
105. Assembly incorrect.
106. Adapters and caps not as specified.
107. Identification marking missing, incomplete, or illegible.
108. Treatment and painting not as specified.
109. Weights not as specified.
110. Workmanship not as specified.

4.5.2 Tests. All test data shall be corrected to standard sea level atmospheric conditions of 29.92 inches of mercury and the specific weight of water of 62.3 lb/cu ft at a temperature of 68° F. Hoses or pipes connected to the pump shall be the same size as the specified adapters.

4.5.2.1 Hydrostatic pressure. Subject the pump body (including air chamber) and valves to an internal hydrostatic pressure of not less than 110 psi for a period of 2 minutes. Nonconformance to 3.7 shall constitute failure of this test.

4.5.2.2 Performance. Operate the pump at a static suction lift equivalent to 20 feet at standard conditions (see 3.14) for not more than 100 strokes. Failure of the pump to prime itself shall constitute failure of this test. After priming, operate the pump at a rate of not

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less than 25 strokes per minute (each cycle shall consist of two strokes). Determine the capacity of the pump in gpc at the applicable discharge head conditions. Repeat this determination two additional times; the average of the three shall be considered the capacity. Inability to deliver at least 0.60 gpc, plus or minus 0.03 gpc, or 1.30 gpc, plus or minus 0.06 gpc, as applicable, shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following major defects. AQL shall be 2.5 percent defective.

111.	Materials, methods, and containers not as specified for Level A or Level B. Each incorrect material, method, or container shall constitute one defect.
112.	Disassembly not as specified for Level A.
113.	Unpainted exterior metal surfaces not coated with preservative as specified for Level A.
114.	Interior surfaces of each pump not preserved as specified, the caps not installed, the piston rods not retracted and secured, and the exposed portion of the piston rod not coated with preservative as specified for Level A.
115.	Repair parts not preserved in accordance with the referenced document for Level A.
116.	Maintenance tools not preserved as specified for Level A.
117.	Consolidation not as specified for Level A.
118.	Packing not as specified for Level A.
119.	Strapping not as specified for Level A.
120.	Marking missing, illegible, incorrect, or incomplete.

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5. PACKAGING

5.1 Preservation. Preservation shall be Level A or Commercial, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Disassembly. The handles shall be removed. Any additional disassembly shall be the minimum necessary to accomplish reduction in cube in accordance with the contractor's recommendation. Removed bolts, nuts, and washers shall be reinstalled in one of the mating parts and secured to prevent loss.

5.1.1.2 Preservatives. Preservatives specified herein shall conform to the applicable specifications listed in and shall be applied in accordance with MIL-P-116.

5.1.1.3 Unprotected exterior metal surfaces. Unpainted exterior metal surfaces of the pumps requiring the application of a contact preservative in accordance with MIL-P-116 shall be coated with Type P-1 preservative.

5.1.1.4 Pumps. The interior surfaces of each pump shall be coated with Type P-7 or P-10 preservative, Type I, Grade 30, applied by fogging through the suction and discharge openings while the pump is being actuated. Excess preservative shall be drained and the openings sealed with the caps provided. The piston rod shall be retracted and secured to prevent movement and the exposed unprotected portion of the piston rod shall be coated with Type P-1 preservative.

5.1.1.5 Repair parts. The preservative application criteria and applicable method(s) of preservation of MIL-P-116 shall be used to preserve repair parts.

5.1.1.6 Maintenance tools. Maintenance tools shall be preserved in accordance with the Level A requirements of PPP-P-40.

5.1.1.7 Technical publications. Technical publications shall be preserved in accordance with MIL-P-116, Method IC-1 or IC-3.

5.1.1.8 Consolidation. The repair parts, maintenance tools, and any small components removed (not including the handles) shall be consolidated in a close-fitting box conforming to PPP-B-636, W6c or W5c, as applicable to the weight.

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5.1.2 Commercial. Each pump, unassembled as recommended by the contractor, and the repair parts and tools shall be preserved and packaged in a manner to assure protection against deterioration and damage from the contractor to the initial destination.

5.2 Packing. Packing shall be Level A, Level B, or Commercial, as specified (see 6.2).

5.2.1 Level A. Each complete pump, preserved as specified in 5.1, shall be packed in a close-fitting box conforming to PPP-B-601, Grade B, Overseas Type, style optional. The contents shall be blocked, braced, and anchored within the box in a manner to prevent movement or damage. Strapping shall conform to QQ-S-781, Class 1, Type I or IV, size as applicable. Unless otherwise specified (see 6.2), strapping shall be Finish B. When specified (see 6.2), strapping shall be Finish A.

5.2.2 Level B. Each complete pump, preserved as specified in 5.2.1, except the box, shall be Domestic Type and strapping shall be either Finish A or Finish B.

5.2.3 Commercial. Each complete pump, preserved as specified in 5.1, shall be packed to assure carrier acceptance and safe delivery to destination at lowest ratings in containers complying with Uniform Freight Classification rules or National Motor Freight Classification.

5.3 Marking.

5.3.1 Military. Military marking shall be in accordance with MIL-STD-129.

5.3.2 Commercial. Commercial marking shall reflect as a minimum, the item name, national stock number, quantity, unit of issue, contract or purchase order number, and appropriate address markings. The marking shall be legible and shall provide a definite contrast.

6. NOTES

6.1 Intended use. The pumps are intended for general service use for transferring lubricating oil, fuel oil, and water.

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6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type and size of pump required (see 1.2).
(c) Time frame required for submission of preproduction model, and number of pumps required (see 3.2).
(d) Color required (see 3.16).
(e) Degree of preservation and degree of packing required (see 5.1 and 5.2).
(f) When strapping shall be other than Finish B (see 5.2.1).

6.3 Preproduction model. Any changes or deviations of production pumps from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the contractor of his obligation to furnish pumps conforming to this specification.

6.4 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operation manual to be furnished with each pump.

6.5 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools, as necessary (including any special tools), and instructions regarding shipment of pump units.

Custodian:
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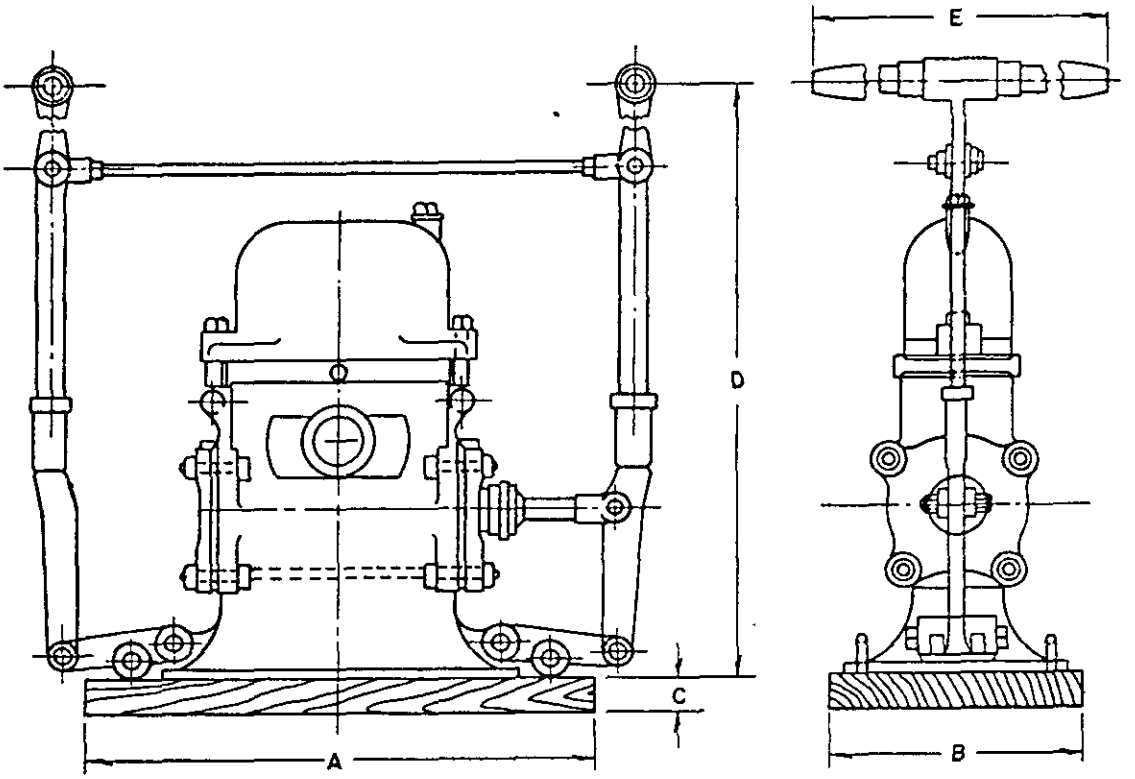
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SIDE ELEVATION

END ELEVATION

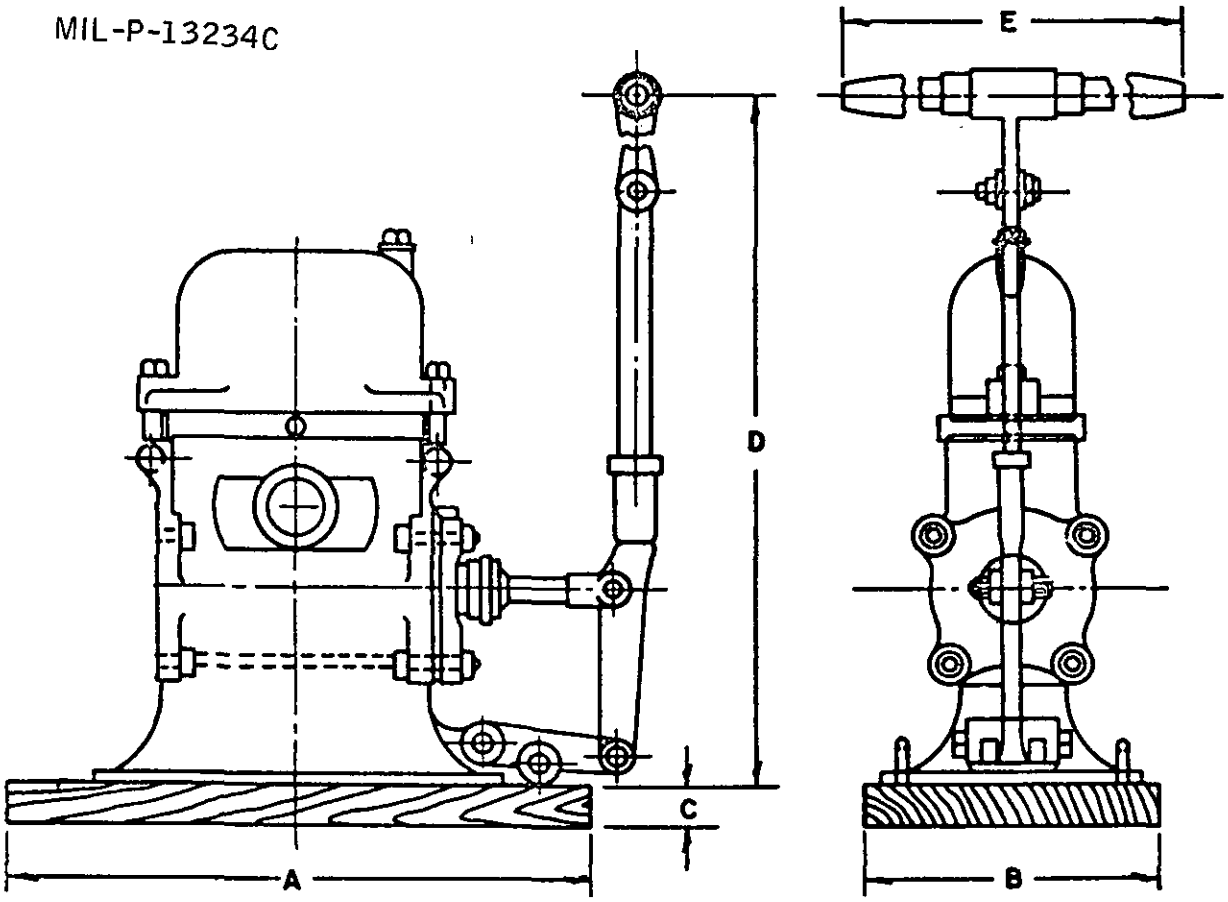
Note:
All dimensions are in inches

Dimensions	0.60Gal.-per-cycle pump	1.30 Gal.-per-cycle pump
A	24 ±2	28±2
B	12 ± 1	14 ± 1
C	2±1/4	
D	40±3	
E	26±1	

FIGURE 1. Double Lever

X-2060

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SIDE ELEVATION

END ELEVATION

Note:
All dimensions are in inches.

Dimensions	0.60 Gal-per-cycle pump
A	24±2
B	12±1
C	2 ± 1/4
D	40±3
E	26±1

FIGURE 2. Single Lever

X-2061

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DOCUMENT IDENTIFIER AND TITLE

MIL-P-13234C Pumps, Reciprocating, Hand-Driven

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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